MSG No. 127.398 P_I4/mbm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,6,20,21,23,24]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,5,7,8,17,18,19,22]`
3	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,10,11,14,28,29,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[12,13,15,16,25,26,27,30]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,15,16,20,21,26,27]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,12,13,23,24,25,30]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[4,5,10,11,17,22,31,32]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,8,9,14,18,19,28,29]`

Table 3: Wyckoff site: `4c`, site symmetry: `4/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,6,9,10,11,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,5,7,8,12,13,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,18,19,22,25,26,27,30]`
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[20,21,23,24,28,29,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,6,7,8,9,14,15,16]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,4,5,10,11,12,13]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[17,22,23,24,25,30,31,32]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[18,19,20,21,26,27,28,29]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,15,23,25]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,13,21,26]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,12,20,27]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,11,19,28]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,10,18,29]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[6,16,24,30]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[7,9,17,31]`

8	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	[8,14,22,32]

Table 6: Wyckoff site: 8f, site symmetry: 4..

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,2,3,6]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[4,5,7,8]
3	$[0,\ 0,\ -z]$	[9,10,11,14]
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[12,13,15,16]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[17,18,19,22]
6	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[20,21,23,24]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[25,26,27,30]
8	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[28,29,31,32]

Table 7: Wyckoff site: 8g, site symmetry: 2.mm

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,6,15,16]
2	$\left[\frac{1}{2},\ 0,\ z\right]$	[2,3,12,13]
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	[4,5,10,11]
4	$\left[0,\ \frac{1}{2},\ -z\right]$	[7,8,9,14]
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[17,22,31,32]
6	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[18,19,28,29]
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[20,21,26,27]
8	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[23,24,25,30]

Table 8: Wyckoff site: 8h, site symmetry: m.2m

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	[1,7,14,16]
2	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	[2,5,11,12]
3	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	[3,4,10,13]
4	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	[6,8,9,15]
5	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[17,23,30,32]
6	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,21,27,28]
7	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[19,20,26,29]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	[22,24,25,31]

Table 9: Wyckoff site: 16i, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	[1,23]
2	$\left[-x,\ x,\ \frac{1}{4}\right]$	[2,21]
3	$\left[x,\ -x,\ \frac{1}{4}\right]$	[3,20]

continued ...

Table 9

No.	position	mapping
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[4,19]
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[5,18]
6	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[6,24]
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[7,17]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[8,22]
9	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[9,31]
10	$\left[x,\ -x,\ \frac{3}{4}\right]$	[10,29]
11	$\left[-x,\ x,\ \frac{3}{4}\right]$	[11,28]
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[12,27]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[13,26]
14	$\left[x,\ x,\ \frac{3}{4}\right]$	[14,32]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[15,25]
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[16,30]

Table 10: Wyckoff site: 16j, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,20]
2	$\left[0,\ x,\ \frac{1}{4}\right]$	[2,23]
3	$\left[0,\ -x,\ \frac{1}{4}\right]$	[3,24]
4	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[4,17]
5	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,22]
6	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[6,21]
7	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[7,18]
8	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[8,19]
9	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[9,28]
10	$\left[0,\ -x,\ \frac{3}{4}\right]$	[10,31]
11	$\left[0,\ x,\ \frac{3}{4}\right]$	[11,32]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[12,25]
13	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[13,30]
14	$\left[x,\ 0,\ \frac{3}{4}\right]$	[14,29]
15	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[15,26]
16	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[16,27]

Table 11: Wyckoff site: 16k, site symmetry: m..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$[-y,\ x,\ 0]$	[2,11]
3	$[y,\ -x,\ 0]$	[3,10]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[4,13]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[5,12]
6	$[-x,\ -y,\ 0]$	[6,9]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[7,16]

continued ...

Table 11

No.	position	mapping
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	[8,15]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[17,30]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,27]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[19,26]
12	$\left[x,\ -y,\ \frac{1}{2}\right]$	[20,29]
13	$\left[-x,\ y,\ \frac{1}{2}\right]$	[21,28]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[22,25]
15	$\left[y,\ x,\ \frac{1}{2}\right]$	[23,32]
16	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[24,31]

Table 12: Wyckoff site: 16l, site symmetry: ..m

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,16]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,12]
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[3,13]
4	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[4,10]
5	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[5,11]
6	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[6,15]
7	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[7,14]
8	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	[8,9]
9	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[17,32]
10	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[18,28]
11	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[19,29]
12	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[20,26]
13	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[21,27]
14	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	[22,31]
15	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[23,30]
16	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[24,25]

Table 13: Wyckoff site: 32m, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[4]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[7]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$[y,\ -x,\ -z]$	[10]
11	$[-y,\ x,\ -z]$	[11]

continued ...

Table 13

No.	position	mapping
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[12]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[23]
24	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27]
28	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[30]
31	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[31]
32	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[32]